United States Patent Application Publication	20250282272
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Standifer; Cliff

TRUCK BED MOUNTED LIFTING DEVICE

Abstract

A hoist assembly for a vehicle cargo bed may include a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed, a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state, a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state, and a carriage assembly operably coupling the first pivot arm to the base portion and enabling the first pivot arm to move relative to the base portion along a longitudinal axis of the base portion while the first pivot arm is in the deployed state.

Inventors: **Standifer; Cliff (Canton, MI)**

Applicant: **Ford Global Technologies, LLC** (Dearborn, MI)

Family ID: **96810522**

Appl. No.: **18/597026**

Filed: **March 06, 2024**

Publication Classification

Int. Cl.: **B60P1/54** (20060101)

U.S. Cl.:

CPC **B60P1/5461** (20130101);

Background/Summary

TECHNICAL FIELD
[0001] Example embodiments generally relate to vehicle accessories and, more particularly, relate to a lifting device mounted in the bed of a truck.
BACKGROUND
[0002] Pickup trucks, which are often considered to be light duty trucks, are some of the most popular vehicles on the road. These trucks, and their related medium and heavy duty cousins, are popular at least in part because the open beds that they commonly possess provide a great deal of flexibility for operators with respect to payloads that can be carried. That said, the addition of accessories, as either a dealer option or an aftermarket add on, gives operators even more flexibility to customize or expand the capabilities of their vehicles, and further fuel the popularity of these vehicles.
[0003] One such accessory may be a truck bed hoist. Since the hoist must necessarily take up space within the truck bed, the balance between size and capability may be difficult to achieve. Example embodiments aim to provide a highly capable truck bed hoist that still minimizes its footprint within the truck bed.
BRIEF SUMMARY OF SOME EXAMPLES
[0004] In accordance with an example embodiment, a hoist assembly for a vehicle cargo bed may be provided. The hoist assembly may include a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed, a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state, a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state, and a carriage assembly operably coupling the first pivot arm to the base portion and enabling the first pivot arm to move relative to the base portion along a longitudinal axis of the base portion while the first pivot arm is in the deployed state.
[0005] In another example embodiment, a hoist assembly for a vehicle cargo bed may be provided, and include a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed. The hoist assembly may also include a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state, and a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state. The first pivot arm may be extended along a deployment axis to be substantially perpendicular to the base portion in the deployed state and folded to be substantially parallel to the base portion in the storage state. The first pivot arm may also be both moveable in a direction perpendicular to the deployment axis and rotatable about the deployment axis in the deployed state.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)
[0006] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
[0007] FIG. **1** illustrates a perspective view of a vehicle cargo bed with a hoist assembly in accordance with an example embodiment;
[0008] FIG. **2** illustrates a perspective view of the hoist assembly in isolation in accordance with an example embodiment;
[0009] FIG. **3** illustrates an exploded view of components of the hoist assembly in accordance with an example embodiment;
[0010] FIG. **4**A illustrates a side view of a winch assembly in accordance with an example

embodiment;
[0011] FIG. **4**B illustrates a perspective view of the winch assembly of FIG. **4**A in accordance with an example embodiment;
[0012] FIG. **5**A illustrates a perspective view of a tower casing in accordance with an example embodiment;
[0013] FIG. **5**B illustrates a cross section view of the tower casing in accordance with an example embodiment;
[0014] FIG. **6** illustrates deployment of a first pivot arm while transitioning toward a deployed state of the hoist assembly in accordance with an example embodiment;
[0015] FIG. **7** illustrates a cross section view of the hoist assembly after deployment of the first pivot arm while transitioning toward the deployed state of the hoist assembly in accordance with an example embodiment;
[0016] FIG. **8** illustrates locking of the first pivot arm upright while transitioning toward the deployed state of the hoist assembly in accordance with an example embodiment;
[0017] FIG. **9**A illustrates deployment of a second pivot arm while transitioning toward the deployed state of the hoist assembly in accordance with an example embodiment;
[0018] FIG. **9**B illustrates the second pivot arm being fully deployed while transitioning toward a deployed state of the hoist assembly in accordance with an example embodiment; and
[0019] FIG. **10** illustrates a perspective view of the hoist assembly in the deployed state in accordance with an example embodiment.

DETAILED DESCRIPTION

[0020] Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.
[0021] Some example embodiments described herein may provide a truck bed hoist that is sized to take up a relatively small area within the truck bed, but still provides enhanced capability due at least in part to its rigid mounting to the chassis (e.g., frame or sub frame) of the vehicle (instead of merely being mounted to the floor of the bed). While maintaining a strong foundation due to its rigid mounting to the chassis, the truck bed hoist is foldable into a relatively small storage position, and extendable into a deployed position that allows both extension and rotation that give the truck bed hoist extensive reach and capability for moving objects to and within the truck bed from beyond an extended tail gate.
[0022] FIG. **1** illustrates a perspective view of a hoist assembly **100** in a storage state within a vehicle cargo bed **110**. The vehicle cargo bed **110** may be defined by a floor **112**, three sidewalls **114** (only two of which are shown in FIG. **1** to enhance visibility of the hoist assembly), and a tail gate **116**. The tail gate **116** may be extended as shown in FIG. **1** and lie parallel with the floor **112** when extended. However, when the tail gate **116** is closed, the tail gate **116** essentially forms a fourth sidewall enclosing the vehicle cargo bed **110**.
[0023] The vehicle may include a chassis formed from an assembly of frame members or subframe members. However, in some cases, the chassis may include a cast frame. In the example of FIG. **1**, the chassis may include a first subframe member **120** and a second subframe member **122** that may extend substantially longitudinally along left and right sides of the vehicle, respectively. Unlike conventional truck bed hoists, which may be mounted to the floor **112** (if mounted at all), example embodiments may mount the hoist assembly **100** to one of the first subframe member **120** or the

second subframe member **122** through floor openings **118** that are formed in the floor **112** and extend through the floor **112** to provide access for a fastener (e.g., a threaded fastener) to attach the hoist assembly **100** either directly or indirectly to the first subframe member **120** or the second subframe member **122**. When such connection is direct, the threaded fastener may pass through a portion of the hoist assembly **100**, and through the floor opening **118** to thread into a receptacle formed at the first subframe member **120** or the second subframe member **122**. When such connection is indirect, a bracket may be attached to the first subframe member **120** and the second subframe member **122** proximate to each instance of the floor opening **118**, and the threaded fastener may pass through the portion of the hoist assembly **100**, and through the floor opening **118** to thread into the bracket attached to the first subframe member **120** or the second subframe member **122**.

[0024] Notably, the first subframe member **120** and the second subframe member **122** extend along sides of the vehicle cargo bed **110** (and therefore parallel to and relatively close to the sidewalls **114** that are perpendicular to the tail gate **116** (when closed). The mounting of the hoist assembly **100** to the first and second subframe members **120** and **122** therefore by necessity places the hoist assembly **100** off to one side or the other of the vehicle cargo bed **110** thereby maximizing the available space alongside the hoist assembly **100** into which an object hoisted by the hoist assembly **100** can be located. The positioning of the hoist assembly **100** is therefore strategic both in terms of space efficiency and robust mounting. Moreover, as will be discussed in greater detail below, the structure and componentry employed in the hoist assembly **100** is further designed to maximize space efficiency and robust capability within the unique environment of the vehicle cargo bed **110** by enabling the hoist assembly **100** to remain fully assembled, but easily capable of transitioning between the storage state shown in FIG. **1** and a deployed state during which the hoist assembly **100** is used to hoist an object. The deployment can therefore transition the hoist assembly **100** from the storage state in which the hoist assembly **100** is folded to ensure no extension above the sidewalls **114** so that operator visibility remains completely unobscured, to the deployed state where full utility of the hoist assembly **100** can be achieved without any disassembly or reassembly.

[0025] Turning to FIGS. **2** and **3**, some major subassemblies or components of the hoist assembly **100** will be described in greater detail. In this regard, FIG. **2** shows a perspective view of the hoist assembly **100** in an assembled condition while in the storage state, and FIG. **3** shows a partially exploded view of various subassemblies or components of the hoist assembly **100**. As shown in FIGS. **2** and **3**, the hoist assembly **100** may include a base portion **200** and a carriage assembly **210** that is designed to be movable relative to the base portion **200**, but otherwise be retained in slidable contact with the base portion **200** via a wheel or roller assembly that will be discussed in greater detail below.

[0026] The base portion **200** may include a mounting plate **202** that lies flat against the floor **112**. The mounting plate **202** may include mounting holes **204** that may be spaced apart by sufficient distance to permit alignment with the floor openings **118**. A set of retention grooves **206** may be formed on the mounting plate **202** to extend substantially parallel to a longitudinal axis **208** of the base portion **200**. The retention grooves **206** of this example may be formed as C channels that extend along an entire length of the mounting plate **202** in the direction of the longitudinal axis **208** and are closed at each opposing longitudinal end thereof. Meanwhile the retention grooves **206** are open in the directions facing each other so that rollers **212** of the carriage assembly **210** can ride in the retention grooves **206** to reposition the carriage assembly **210** along the longitudinal axis **208**. The rollers **212** are spaced apart from each other along the longitudinal axis **208** by a distance shorter than the length of the retention grooves **206** and the closure of the ends of the retention grooves **206** limits the range of motion of the rollers **212** accordingly.

[0027] The carriage assembly **210** may include a carriage body **214** that includes plate members defining sides of the carriage assembly **210** to which various other components may be attached. In this regard, for example, the carriage assembly **210** may be operably coupled to a first pivot arm **220** that pivots relative to the carriage assembly **210** to transition between the storage state and the deployed state. To accomplish the pivoting action, the first pivot arm **220** may connect to the carriage assembly **210** at one fixed pivot location, and at one variable support location. In an example embodiment, the first pivot arm **220** may include a tower casing **222**, and the tower casing **222** may host components for connection forming both the fixed pivot location and the variable support location. In this regard, the tower casing **222** may include a set of pivot bosses **224** disposed opposite each other at a bottom end of the tower casing **222**. The pivot bosses **224** may pivotally connect to opposite sidewalls of the carriage body **214** to define a first pivot axis **226** about which the tower casing **222** pivots to transition between the deployed state and the storage state. The variable support location is defined by a slide assembly **216** formed on internal sides of the carriage body **214** opposite each other and facing each other.

[0028] A slide brace **218** is attached (pivotally) to the tower casing **222** at one end thereof, and slidably fitted into the slide assembly **216** at the other end thereof. When the hoist assembly **100** is transitioned to the deployed state, a lower tube **219** of the slide brace **218** slides within the slide assembly **216** toward the pivot bosses **224**. When the hoist assembly **100** is transitioned to the storage state, the lower tube **219** of the slide brace **218** slides away from the pivot bosses **224** and toward the rollers **212**. In an example embodiment, either one or both opposite ends of the slide assembly **216** may include a detent aimed at retaining the lower tube **219** at either end of the slide assembly **216**. The operator may apply a lifting force or lowering force to the first pivot arm **220** to urge the lower tube **219** past the detent at each opposing end of the slide assembly **216** to transition the hoist assembly **100** between the deployed state and the storage state. However, as will be discussed in greater detail below, further retention means may also be employed to hold the lower tube **219** in place in the deployed state (namely retention pin **290**).

[0029] The carriage assembly **210** may also include a handle **217**, which may be operably coupled to the carriage body **214**. The handle **217** may be used to urge the carriage assembly **210** to slide out over the tail gate **116**, or to urge the carriage assembly **210** to slide back toward the front of the vehicle to permit transition back to the storage state. The handle **217** may be used to slide the carriage assembly **210** either when the first pivot arm **220** is folded (e.g., to the position shown in FIGS. **1** and **2**) or when the first pivot arm **220** is extended (e.g., in the position shown in FIGS. **6-10**).

[0030] The first pivot arm **220** is operably coupled to the carriage assembly **210** at a first end thereof by the tower casing **222**, as described above. The tower casing **222** may be considered to be a lower tower assembly by virtue of its position closest to the carriage assembly **210** when the first pivot arm **220** is oriented vertically or upright. The second end of the first pivot arm **220** is operably coupled to a second pivot arm **230** at an upper tower assembly **240**, so named due to its being farther away from the carriage assembly **210** and above the lower tower assembly when the first pivot arm **220** is oriented vertically or upright. The upper tower assembly **240** may include a first plate assembly **242** providing a pivotal connection with a proximal end of the second pivot arm **230**. The first plate assembly **242** may include a set of plates or brackets attached to opposite sides of a shaft portion **244** of the first pivot arm **220** to allow connection to other components. A pin, shaft, or other cylindrical component may pass through the first plate assembly **242** and the proximal end of the second pivot arm **230** to allow the second pivot arm **230** to pivot about the pin, shaft or other cylindrical component when transitioning between the deployed state and the storage state.

[0031] The second pivot arm **230** may also include a second plate assembly **246** that provides a pivotal connection with a first link arm **248** at a distal end of the second pivot arm **230**. In this regard, the second plate assembly **246** may include a set of plates or brackets attached to opposite sides of the second pivot arm **230** to allow a pin, shaft, or other cylindrical component to pass through the second plate assembly **246** and the first link arm **248** to enable the first link arm **248** to pivot about the pin, shaft or other cylindrical component when transitioning between the deployed state and the storage state.

[0032] A third plate assembly **250** may be provided at the first pivot arm **220** proximate to the tower casing **222** to retain a pivotal connection with a second link arm **252** at the tower casing **222**. The first and second link arms **248** and **252** may be pivotally attached to each other at ends thereof opposite the second pivot arm **230** and the first pivot arm **220** respectively to allow the first and second link arms **248** and **252** to fold relative to each other (and to both the first and second pivot arms **220** and **230** in the storage position shown in FIGS. **2** and **3**). However, the first and second link arms **248** and **252** may be extended to align with each other between the second and third plate assemblies **246** and **250** to hold the second pivot arm **230** in the deployed state. When the first link arm **248** is extended relative to the second link arm **252**, a latch assembly **251** may be employed to secure both the first and second link arms **248** to ensure that they stay in this position, even in the presence of a load being carried by the hoist assembly **100**.

[0033] The upper tower assembly **240** may include a winch assembly **260**, which may be attached to one plate of the first plate assembly **242**. The winch assembly **260** may be operated to move a lifting member **262** (e.g., a strap, rope, chain or other flexible elongated member) to alternately lift or lower objects attached to the lifting member **262**. In some cases, a distal end of the lifting member **262** may include a hook **270** or other attachment means that can engage the object that is to be lifted using the hoist assembly **100**. The hook **270** may also be used, in some cases and if desired, as a latch or lock to the tower casing **222**. In this regard, for example, the hook **270** may be attached to a portion of the tower casing **222** to lock the second pivot arm **230** to the first pivot arm **220** (e.g., as shown in FIG. **6**) until deployment is desired.

[0034] In an example embodiment, the winch assembly **260** may include a gear assembly including a drive gear **300** and a driven gear **310** that is operably coupled to an output shaft **320** that, when rotated either winds the lifting member **262** onto or off of a spool or reel assembly. The drive gear **300** is mounted on a drive shaft **330**, as shown in FIGS. **4**A and **4**B. The drive shaft **330** of this example is double ended, in that the drive shaft **330** may include a first end **340** and a second end **350**, each of which include a hex interface drivable via a hand crank or a power tool. The drive shaft **330** may extend substantially parallel to the floor **112** of the vehicle cargo bed **110** in the deployed state, but may be perpendicular to the floor **112** in the storage state (as shown in FIG. **1**). Notably, the inclusion of the double ended drive shaft **330** may allow for convenient operation of the drive shaft **330** regardless of the orientation of the second pivot arm **230**. In this regard, since the second pivot arm **230** may rotate 360 degrees, the positioning of the drive shaft **330** relative to the operator may be different in different scenarios, and may change during a given hoisting operation such that it may be convenient to switch between applying turning force to the drive shaft **330** via the first end **340** or the second end **350**.

[0035] To enable the rotation of the second pivot arm **230**, the first pivot arm **220** employs the tower casing **222** to house a first bearing assembly **400** and a second bearing assembly **410** that allow the shaft portion **244** to rotate 360 degrees within the tower casing **222**. FIGS. **5**A and **5**B show a perspective view of the tower casing **222** and a cross section view through the tower casing **222**, respectively. Meanwhile, the shaft portion **244** may be retained in the tower casing **222** by a castle nut **420** and cotter pin **430** separated from the second bearing assembly **410** by a spacer tube **440**.

[0036] Operation of the hoist assembly **100** in transition from the storage state to the deployed state will now be discussed in reference to FIGS. **2**, **3** and **6-10**. As shown in FIG. **2**, the retention pin **290** (which may also be referred to as a locking pin) may be used to affix the first pivot arm **220** to the carriage assembly **210**. More particularly, in this example, the retention pin **290** may attach the carriage body **214** to the first plate assembly **242**. By removing the retention pin **290**, the first pivot arm **220** may be enabled to transition from the storage state, in which the first pivot arm **220** lies substantially parallel to the floor **112** (as shown in FIG. **1**), to an upright condition shown in FIG. **6**.

This upright condition extends the first pivot arm **220** along a deployment axis **500** that is substantially perpendicular to the base portion **200**. The transition may occur responsive to the operator pivoting the first pivot arm **220** about the pivot axis **226** of the tower casing **222** in the direction shown by arrow **510** in FIG. **6**.

[0037] Referring now also to FIG. **7**, the slide brace **218** moves through the slide assembly **216** in the direction of arrow **520** to reach the position shown in FIG. **7**. At this stage, the second pivot arm **230** may remain parallel to the first pivot arm **220** (and the deployment axis **500**), and substantially perpendicular to the base portion **200** and the floor **112**. Thereafter, a locking orifice **530** formed in the carriage body **214** may be aligned with a central opening in the lower tube **219** of the slide brace **218** and the same pin (i.e., the locking pin **290**) that locked the first pivot arm **220** in the storage state may be used to lock the first pivot arm **220** in the deployed state as shown in FIG. **8** due to insertion of the locking pin **290** in the direction of arrow **540**.

[0038] Thereafter, as shown in FIG. **9**A, the first and second link arms **248** and **252** may be extended away from the second pivot arm **230** and the first pivot arm **220** (and each other) as shown by arrows **550**. This pivoting begins to extend the second pivot arm **230** away from the first pivot arm **220** as shown by arrow **560**. When the first and second link arms **248** and **252** are aligned with each other, and locked in such alignment, the second pivot arm **230** may be fully extended and ready for operation as shown in FIG. **9**B. If desired (although not required), the carriage assembly **210** may be extended in the direction of arrow **570** shown in FIG. **10**. The first and second bearing assemblies **400** and **410** of FIG. **5**B may then allow the first pivot arm **220** to rotate 360 degrees about the deployment axis **500** in the directions of arrow **580** as shown in FIG. **10**. However, it should be noted that hoisting operations may be conducted either with the carriage assembly **210** slid out over the tail gate **116**, or without sliding the carriage assembly **210** at all. Moreover, the hoist assembly **100** could also be operated with the tail gate **116** closed. Stowage of the hoist assembly **100** may be conducted via a reversal of the operations outlined above. In some embodiments, the interface between the tower casing **222** and a lower portion of the first pivot arm **220** may employ one or more instances of a detent **581** or other rotation inhibitor, which may hold or assist in holding the first pivot arm **220** in a given position relative to the rotation about the deployment axis **500**.

[0039] A hoist assembly for a vehicle cargo bed may be therefore be provided in accordance with an example embodiment. The hoist assembly may include a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed, a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state, a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state, and a carriage assembly operably coupling the first pivot arm to the base portion and enabling the first pivot arm to move relative to the base portion along a longitudinal axis of the base portion while the first pivot arm is in the deployed state.

[0040] The hoist assembly (or a vehicle including the same) of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the device. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the carriage assembly may be entirely within the vehicle cargo bed in the storage state, and may enable the first pivot arm to slide over a portion of an extended tail gate in the deployed state. In an example embodiment, the first pivot arm may be substantially perpendicular to a floor of the vehicle cargo bed in the deployed state and substantially parallel to the floor in the storage state. In some cases, the base portion may attach to the subframe via a threaded fastener that extends through the floor of the vehicle cargo bed. In an example embodiment, the first pivot arm may include a lower tower assembly disposed at the first end of the first pivot arm. The lower tower assembly may include a tower casing comprising a pivot boss pivotally operably coupling the first pivot arm to the carriage assembly at a bottom end of the tower casing, and the tower casing may include a pivot bracket operably coupling a top end of the tower casing to a slide brace. The slide brace may slide within a slide assembly formed in the carriage assembly when transitioning between the deployed state and the storage state. In some cases, the tower casing may further include a bearing assembly enabling a tower shaft of the first pivot arm to rotate 360 degrees within the tower casing when the first pivot arm is in the deployed state. In an example embodiment, the first pivot arm may include an upper tower assembly disposed at the second end of the first pivot arm. The upper tower assembly may include a first plate assembly retaining a pivotal connection with a proximal end of the second pivot arm, and a second plate assembly retaining a pivotal connection with a first link arm at a distal end of the second pivot arm. A third plate assembly may retain a pivotal connection with a second link arm at the tower casing, and the first and second link arms may extend in alignment with each other between the second and third plate assemblies to hold the second pivot arm in the deployed state. In some cases, the upper tower assembly may operably couple a winch assembly to the hoist assembly to operate a lifting member to alternately lift or lower objects attached to the lifting member. In an example embodiment, the winch assembly may include a gear assembly operably coupled to a drive shaft, and the drive shaft may include a first end and a second end, each of which include a hex interface drivable via a hand crank or a power tool. In some cases, the drive shaft may extend substantially parallel to a floor of the vehicle cargo bed in the deployed state. In an example embodiment, a locking pin locks the upper tower assembly to the base portion in the storage state, and the locking pin locks the slide brace at a selected position of the slide assembly in the deployed state. In some cases, the first pivot arm may be locked in both the deployed state and the storage state by a same locking pin disposed at two different locking locations. In an example embodiment, the first pivot arm may extend higher than sidewalls of the vehicle cargo bed in the deployed state, and an entirety of the second pivot arm may be higher than the sidewalls in the deployed state. In this context, neither the first pivot arm nor the second pivot arm may extend higher than the sidewalls in the storage state. In some cases, the first and second pivot arms remain operably coupled to each other, and the first pivot arm remains operably coupled to the base portion via the carriage assembly during vehicle operation in the storage state. In an example embodiment, the first pivot arm may be extended along a deployment axis to be substantially perpendicular to the base portion in the deployed state and folded to be substantially parallel to the base portion in the storage state, and the first pivot arm may be both movable in a direction perpendicular to the deployment axis and rotatable about the deployment axis in the deployed state. In some cases, the base portion may include a pair of retention grooves, and the carriage assembly may include a set of rollers that ride within the retention grooves to carry the first pivot arm along the base portion in the deployed state.

[0041] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A hoist assembly for a vehicle cargo bed, the hoist assembly comprising: a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed; a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state; a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state; and a carriage assembly operably coupling the first pivot arm to the base portion and enabling the first pivot arm to move relative to the base portion along a longitudinal axis of the base portion while the first pivot arm is in the deployed state.
2. The hoist assembly of claim 1, wherein the carriage assembly is entirely within the vehicle cargo bed in the storage state, and enables the first pivot arm to slide over a portion of an extended tail gate in the deployed state.
3. The hoist assembly of claim 1, wherein the first pivot arm is substantially perpendicular to a floor of the vehicle cargo bed in the deployed state and substantially parallel to the floor in the storage state.
4. The hoist assembly of claim 3, wherein the base portion attaches to the subframe via a threaded fastener that extends through the floor of the vehicle cargo bed.
5. The hoist assembly of claim 1, wherein the first pivot arm comprises a lower tower assembly disposed at the first end of the first pivot arm, the lower tower assembly comprising a tower casing comprising a pivot boss pivotally operably coupling the first pivot arm to the carriage assembly at a bottom end of the tower casing, and wherein the tower casing comprises a pivot bracket operably coupling a top end of the tower casing to a slide brace, the slide brace sliding within a slide assembly formed in the carriage assembly when transitioning between the deployed state and the storage state.
6. The hoist assembly of claim 5, wherein the tower casing further comprises a bearing assembly enabling a tower shaft of the first pivot arm to rotate 360 degrees within the tower casing when the first pivot arm is in the deployed state.
7. The hoist assembly of claim 5, wherein the first pivot arm comprises an upper tower assembly disposed at the second end of the first pivot arm, the upper tower assembly comprising a first plate assembly retaining a pivotal connection with a proximal end of the second pivot arm, and a second plate assembly retaining a pivotal connection with a first link arm at a distal end of the second pivot arm, wherein a third plate assembly retains a pivotal connection with a second link arm at the tower casing, and wherein the first and second link arms extend in alignment with each other between the second and third plate assemblies to hold the second pivot arm in the deployed state.
8. The hoist assembly of claim 7, wherein the upper tower assembly operably couples a winch assembly to the hoist assembly to operate a lifting member to alternately lift or lower objects attached to the lifting member.
9. The hoist assembly of claim 8, wherein the winch assembly comprises a gear assembly operably coupled to a drive shaft, and wherein the drive shaft comprises a first end and a second end, each of which include a hex interface drivable via a hand crank or a power tool.
10. The hoist assembly of claim 9, wherein the drive shaft extends substantially parallel to a floor of the vehicle cargo bed in the deployed state.
11. The hoist assembly of claim 7, wherein a locking pin locks the upper tower assembly to the

base portion in the storage state, and wherein the locking pin locks the slide brace at a selected position of the slide assembly in the deployed state.

12. The hoist assembly of claim 1, wherein the first pivot arm is locked in both the deployed state and the storage state by a same locking pin disposed at two different locking locations.

13. The hoist assembly of claim 1, wherein the first pivot arm extends higher than sidewalls of the vehicle cargo bed in the deployed state, and an entirety of the second pivot arm is higher than the sidewalls in the deployed state, and wherein neither the first pivot arm nor the second pivot arm extends higher than the sidewalls in the storage state.

14. The hoist assembly of claim 1, wherein the first and second pivot arms remain operably coupled to each other, and the first pivot arm remains operably coupled to the base portion via the carriage assembly during vehicle operation in the storage state.

15. The hoist assembly of claim 1, wherein the first pivot arm is extended along a deployment axis to be substantially perpendicular to the base portion in the deployed state and folded to be substantially parallel to the base portion in the storage state, and wherein the first pivot arm is both movable in a direction perpendicular to the deployment axis and rotatable about the deployment axis in the deployed state.

16. The hoist assembly of claim 1, wherein the base portion comprises a pair of retention grooves, and wherein the carriage assembly comprises a set of rollers that ride within the retention grooves to carry the first pivot arm along the base portion in the deployed state.

17. A hoist assembly for a vehicle cargo bed, the hoist assembly comprising: a base portion operably coupled to a subframe of the vehicle in the vehicle cargo bed; a first pivot arm operably coupled to the base portion proximate a first end of the first pivot arm to pivot relative to the base portion between a deployed state and a storage state; and a second pivot arm operably coupled to a second end the first pivot arm to pivot relative to the second end of the first pivot arm when transitioning between the deployed state and the storage state, wherein the first pivot arm is extended along a deployment axis to be substantially perpendicular to the base portion in the deployed state and folded to be substantially parallel to the base portion in the storage state, and wherein the first pivot arm is both moveable in a direction perpendicular to the deployment axis and rotatable about the deployment axis in the deployed state.

18. The hoist assembly of claim 17, further comprising a carriage assembly operably coupling the first pivot arm to the base portion to carry the first pivot arm while moving relative to the base portion in the direction perpendicular to the deployment axis.

19. The hoist assembly of claim 18, wherein the carriage assembly is entirely within the vehicle cargo bed in the storage state, and enables the first pivot arm to slide over a portion of an extended tail gate in the deployed state.

20. The hoist assembly of claim 17, wherein the base portion comprises a pair of retention grooves, and wherein the carriage assembly comprises a set of rollers that ride within the retention grooves to carry the first pivot arm along the base portion in the deployed state.